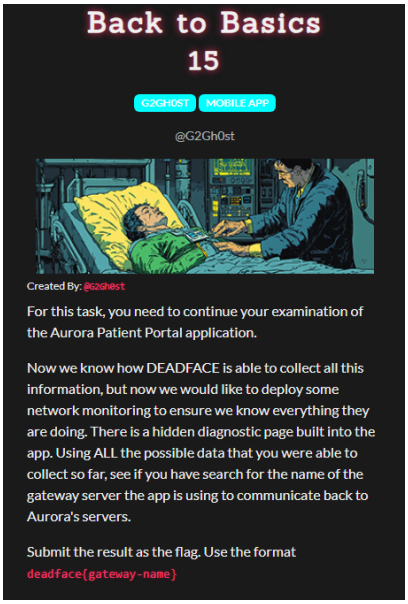
Solving “Back to Basics” (15 pts - Mobile App)

A static-analysis walkthrough for recovering an internal gateway hostname from an iOS application binary.



Challenge page used during the solve.

1. The Challenge

The task asked for the gateway server used by the Aurora Patient Portal and hinted that all necessary information was already in the collected app data. No emulator or debugger was required.

2. Static Analysis

After unpacking the IPA, search printable strings in the main executable. Nearby diagnostic labels expose an internal gateway hostname.

```
CTF patient companion
gw-westlake-int-03
Patient Cache Realm
patient-summary-cache-v2
Medication Cache
offline lookup enabled
```

3. Why This Is the Gateway

The hostname appears in the diagnostic-data region immediately before cache and connectivity status labels, matching the challenge hint about a hidden diagnostic page and gateway communication.

4. The Flag

```
deadface{gw-westlake-int-03}
```

5. General Method

• For mobile CTFs, start with static strings before attempting dynamic instrumentation.

• Inspect the app binary, Info.plist, bundled databases, and resource files.

• Context around a string matters: nearby UI labels often reveal what a value represents.

Summary

Field	Result
Category	Mobile App static analysis
Artifact	iOS IPA executable
Gateway	gw-westlake-int-03
Flag	deadface{gw-westlake-int-03}